The Influence of Positional Tracking Uncertainty on Observed Cascade Onset in a Simplified LEO Simulation

Rishab Sinha

1. ABSTRACT

The increasing population of objects in Low-Earth Orbit (LEO) raises concerns regarding cascading collisions and the long-term use of the orbital environment. This study investigates how positional tracking uncertainty influences whether and when cascade onset is observed in a simplified orbital simulation. A Monte Carlo particle model was developed to maintain separate true and observed histories, allowing Gaussian positional error to affect cascade detection without altering physical motion or collisions. Across 800 simulations, two initial populations were tested at four uncertainty levels. In the present model, cascade onset is operationally identified when any object first reaches the third generation of collision ancestry. Increasing uncertainty substantially raised the observed cascade probability, whereas the corresponding true cascade probability remained unchanged. At positional uncertainty σ = 4 km and initial population N_0 = 6, the observed and true cascade probabilities were 68% and 10%, respectively. Most classification errors were false cascade detections, while matched observed onsets generally appeared earlier than their true counterparts. Under the assumptions of this model, positional uncertainty can therefore distort both the classification and detected timing of cascading behavior.

2. INTRODUCTION

The orbital environment is currently undergoing a paradigm shift as private and public entities accelerate launch activities, particularly in LEO [1], [2]. This rapid expansion has revived critical concerns regarding the Kessler Syndrome, a theoretical threshold where debris densities become sufficient to trigger self-sustaining collisional cascades [3], [4]. A cascade therefore differs from an isolated collision because its resulting debris contributes to later collisions.

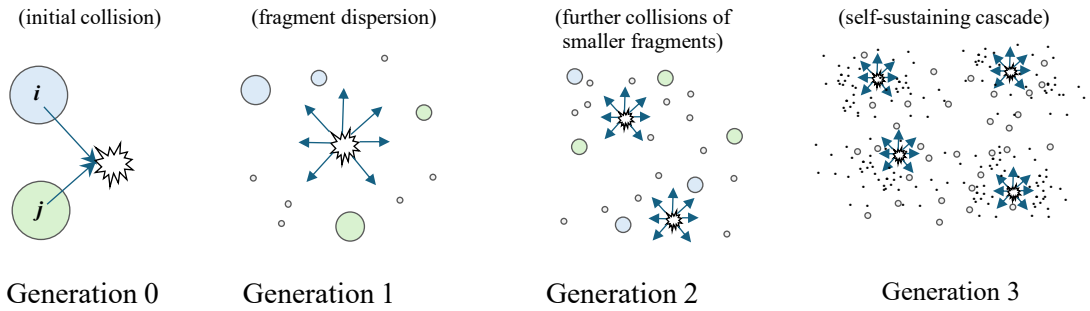


Fig. 1. Conceptual progression of a collisional cascade in this study. Arrows indicate object motion and starbursts indicate collision events. Collision-generated fragments can subsequently collide and produce successive generations of debris. Here, a self-sustaining cascade is represented by fragments continuing to form further collision generations. Generation 3 is deliberately chosen as the cascade onset threshold because it requires multiple generations beyond an isolated collision. Note that it is an operational definition for this simplified model rather than a universal physical threshold.

This severely impacts space-sustainability in the long run [1], [2]. Deterministic models widely study this effect, assuming a perfect position detection, however, real-world Space Situational Awareness (SSA) is fundamentally limited by stochastic tracking uncertainties stemming from sensor noise [5]. Current literature also explores the mechanics of debris multiplication and the statistical quantification of positional error, yet these two domains remain largely siloed.

There is a critical scientific need to connect these concepts to understand how the veracity of positional tracking data affects our ability to detect the onset of a cascade [6]. This study evaluates how positional tracking uncertainty influences the detection and apparent timing of cascading collision behavior in a simplified LEO simulation. Bridging this gap is crucial for real-world collision risk-assessments [7]. These insights can inform both the design of future tracking technologies and the interpretation of observed cascade events, ultimately enhancing space sustainability efforts [5]. Specifically, an isotropic Gaussian error model is used to characterize how expanding spatial uncertainty bubbles can prematurely trigger generation-based cascade thresholds. This mechanism allows quantifying the temporal bias in cascade detection.

3. THE MODEL

The model represents the LEO environment as a dynamic system of individually resolved particles within a two-dimensional space meant to simulate satellites. Each object is characterized by its state vectors: position and velocity. For two arbitrary objects, their relative position and velocity is measured. The system progresses through time using a constant-speed kinematic regime, where the displacement of any object is a linear function of time. To maintain a fixed population density the space employs periodic boundaries, where objects exiting one side of the coordinate plane immediately re-enter from the opposite side.

Two layers are maintained to isolate the influence of tracking uncertainty on determining the time at which the population starts to cascade. The physical layer reflects the truth of whether a particle has collided or not. The observation layer represents the data available to a hypothetical space operator, which displayed an error in apparent particle position. These errors are contrasted with the no-error state (the physical layer) with a confusion matrix to discern how this tracking error may affect real-world satellites.

3.1 Collision Detection

At the beginning of each time step Δt, the simulation analyzes every unique, unordered pair of active objects i and j. For each pair, the model computes the relative displacement vector $\vec{d}$ and the relative velocity vector $\vec{u}$. In this model, $\vec{d}$ is adjusted using the minimum-image convention, in which the displacement is calculated through the nearest periodic copy of an object. The convention selects the shortest displacement to one of these copies. This is to ensure that objects near boundaries are treated as close objects rather than separated by the entire length of the simulation window.

To identify the exact moment when two objects are closest to each other, the algorithm determines the unconstrained time of closest approach t^* by minimizing their squared distance function over continuous time. This yields a projection of the relative displacement onto the relative velocity vector.

$$t^* = -\frac{\vec{d}\cdot\vec{u}}{\vec{u}\cdot\vec{u}} \tag{1}$$

If the objects are moving towards each other, their time of closest approach t^* would be positive. If the objects are moving away from each other, their positive dot product returns a negative t^* value, so the collision occurred before the current time step (which is why it is unconstrained).

$$t_{\mathrm{ca}} = \max\ (0, \min(\Delta t, t^*)) \tag{2}$$

The simulation is evaluated in several time steps of length Δt. Therefore, it defaults a highly positive $t^* > \Delta t$ to the upper limit of the time step Δt , and a negative t^* to the lower limit of zero. Avoiding impossible collision scenarios (i.e. if the relative separation is found to be constant), t_{ca} is also set to zero. This is to ensure simulation efficiency. i.e. only the current time window relevant to measuring collision possibility between two objects is noted as $t_{\mathrm{ca}} \in [0, \Delta t]$.

Then the model evaluates the relative physical separation distance at that exact instant. Over the course of the entire motion of both particles, a point exists when the distance between them reaches a minimum and is defined at t_{ca}. Logically, two particles collide when the distance between them is less than or equal to combined radius of collision (the sum of radii of colliding objects i and j. Consequently, using the Euclidean norm, the straight-line magnitude of the vector is given. The distance of peak proximity d_{min} is given by:

$$d_{\mathrm{min}} = \left\| \vec{\mathrm{d}} + \vec{\mathrm{u}}\mathrm{t}_{\mathrm{ca}} \right\| \tag{3}$$

A Boolean collision indicator C is assigned based on whether the status of the relative distance satisfies this condition at a given instant.

$$\boldsymbol{C} = \begin{cases} 1, & d_{\mathrm{min}} \leq R_i + R_j \\ 0, & d_{\mathrm{min}} > R_i + R_j \end{cases} \tag{4}$$

If the value returns 1 (true), the simulation outputs an alert of a collision (in case of the observation layer, a warning), and removes the colliding parent objects to initiate subsequent particle fragmentation dynamics. Otherwise, the simulation continues unaffected until it receives such a conjunction alert.

3.2 Positional Tracking Uncertainty

To analyze the relationship between tracking error and cascade onset, the simulation implements a stochastic observation model and performance metrics across the system. Every tracked position is modelled as a true physical coordinate corrupted by random sensor noise. The observation layer defines this through an isotropic Gaussian observation model. The random error is centered around zero, with equal independent spread in the horizontal and vertical directions.

$$\vec{\mathrm{r}}_{\mathrm{i,obs}} = \vec{\mathrm{r}}_{\mathrm{i,true}} + \vec{\epsilon}_{\mathrm{i}} \tag{5}$$

$$\text{where, } \vec{\epsilon}_{\mathrm{i,x}}, \vec{\epsilon}_{\mathrm{i,y}} \sim \mathcal{N}(0, \sigma^2) \text{ for each particle } i.$$

This represents the fundamental assumption that the tracking system is unbiased over time (the mean error in both axes is 0) but suffers from a coordinate standard deviation σ. This parameter serves as the independent variable measured across many simulations to determine how badly increasing σ levels degrade situational awareness.

When the observer calculates the distance between two objects, the independent errors of both objects combine, and their variances add. This results in a relative coordinate standard deviation greater than either of the individual values $\sigma = \sigma_x = \sigma_y$ by 41.4%, which increases the misclassification of close approaches in the simulation.

$$\sigma_{rel} = \sqrt{2}\sigma \tag{6}$$

Furthermore, while individual coordinate errors follow a Gaussian distribution, the total distance an object appears from its true location is non-negative and follows a Rayleigh distribution.

$$P(R_\varepsilon \leq r) = 1 - e^{\left(\frac{-r^2}{2\sigma^2}\right)} \tag{7}$$

$$r_P = \sigma\sqrt{-2\,ln(1-p)} \Rightarrow r_{95} \approx 2.448\sigma$$

This allows characterizing the region of uncertainty around a modelled satellite. So, drawing a circle of $R = 2.448\sigma$ around a particle gives a 95% chance that the observed position of it lies within that confidence circle centered on the true position. As σ increases, overlapping error bubbles lead to a higher frequency of false warnings.

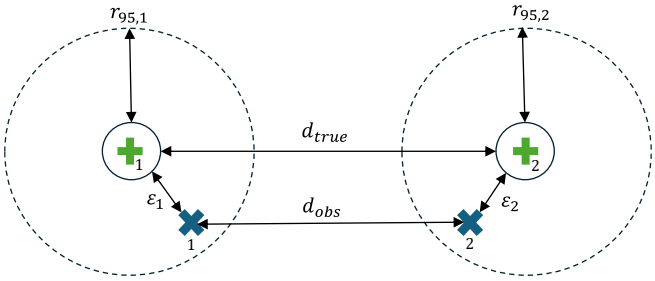


Fig. 2. Representation of positional tracking uncertainty for two objects. Green plus signs denote their true positions, while blue crosses denote the observed positions after independent positional errors $\overrightarrow{\varepsilon_1}$ and $\overrightarrow{\varepsilon_2}$ are applied. d_{true} and d_{obs} represent the separations between the true and observed position pairs, respectively. Dashed circles indicate the 95% positional uncertainty regions, $r_{95} = 2.448\sigma$.

Temporal distortion also affects conjunction predictions. The error in the predicted time of closest approach is governed by the relation:

$$\delta t = -\frac{\vec{\epsilon}_{rel}\cdot\vec{u}}{\|\vec{u}\|^2} \tag{8}$$

If the error component points along the direction of motion, the observer erroneously predicts an earlier collision. If it points opposite, the collision appears delayed. The used convention is viewing these metrics from the observer's point of view.

$$\text{Precision} = \frac{TP}{TP+FP}\,,\ \text{Recall} = \frac{TP}{TP+FN} \tag{9}$$

To quantify the effect's performance, raw counts from the confusion matrix were used. Precision was used to measure the extent of accuracy of collision flagging and was prioritized over accuracy as it was misleading. Recall gives the fraction of physically occurring collisions as the correctly predicted ones.

3.3 Particle Cascading

A cascading mechanism models the transition from isolated collisions to a self-sustaining debris environment. This process is governed by rules of population accounting and ancestry generation, which allow defining the onset of Kessler Syndrome in the simulation. In this model, ancestry refers to the collision history (sequence of events) linking each fragment to the parent objects from which it was generated.

To simulate the physical destruction of satellites and the subsequent multiplication of debris, the model updates the active object count at the moment of impact:

$$N(T^+) = N(T^-) - 2 + n_f \tag{10}$$

Here, $N(T^-)$ represents the population of the simulation window an instant before the collision. The two parent objects are removed from the original tracking list. n_f represents the number of fragments generated, which is determined by a size-dependent rule where larger parents produce more child fragments. If $n_f > 2$, the simulation experiences a net increase in object density, creating the positive feedback loop necessary for a cascade.

To measure the depth of the collision chain and define a cascade onset, generations were defined:

$$g_{\text{new}} = 1 + \max(g_i, g_j) \tag{11}$$

Each object in the 2D space carries a generation tag $g = 0$ initially, which is continuously updated throughout the simulation. This ensures that the cascade is defined by the evolutionary history of the debris, rather than just raw collision numbers. The onset of the cascade is defined as the minimum time in which this generation tag reaches a count of 3 or greater for any particle (g_{max}), as the critical level of ancestry. Then, the simulation tracks how the tracking errors distort this timing using B as the onset bias (difference between the two timings).

$$T_{\text{true}} = \min\{T: g_{\text{max,true}}(T) \geq 3\} \tag{12}$$

$$T_{\text{obs}} = \min\{T: g_{\text{max,obs}}(T) \geq 3\} \tag{13}$$

$$\text{B} = T_{\text{obs}} - T_{\text{true}} \tag{14}$$

If $B < 0$, the observed cascade onset occurs earlier than the true cascade onset.

If $B > 0$, the observed cascade onset occurs later than the true cascade onset.

These equations define the model used to distinguish physical and observed cascade behavior. The following section describes how this framework was implemented computationally and evaluated across repeated simulations.

4. METHODS

The simulation was implemented in Python as a Monte Carlo model. Each run followed a changing population of modelled satellites and generated fragments through discrete time steps, while continuous closest-approach calculations were performed within each step to prevent collisions from being missed. True events and observed events were stored as separate histories. To make large experimental batches computationally practical, graphical animation was not used and instead followed a headless approach. Only the required event summaries were retained, and termination conditions prevented uncontrolled fragment growth. Completed conditions were also recorded so interrupted batches could resume without duplication.

4.1 Simulation Structure

The observation window measures 1000×600 km, giving a total area of $600{,}000$ km^2. Initial objects are distributed randomly, assigned an initial constant effective radius of 1.2 km, and given a speed $s_i \in [7{,}8]$ km s^{-1} at random angles θ_i, approximating the orbital speed of real satellites in LEO [8]. Small velocity changes are applied to fragments following collisions based on the average parent velocities. Fragment radii range from 150 m to 350 m. Consequently, object state vectors and total population can change as fragmentation progresses. The corresponding initialization and periodic-boundary equations are provided in the Appendix (A1–A3).

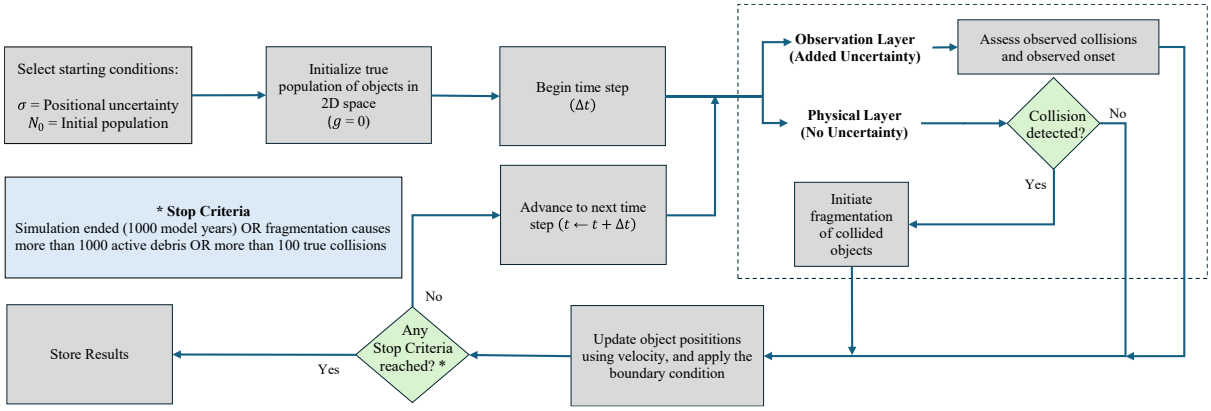


Fig. 3. Computational workflow of the Monte Carlo simulation. The physical layer determines true collision and fragmentation events, while the observation layer independently evaluates uncertainty-perturbed positions to determine observed collisions and observed cascade onset.

The Monte Carlo experiment repeated each selected combination of initial population N_0 and tracking uncertainty σ across many random seeds. Initial tracking parameters and errors therefore varied between repetitions. However, the same seed was paired across different uncertainty levels. Object motion and object error used independent but reasonable random value generation to answer the study's focus. This ensured that changing σ altered only the observed overlaps and not the physical collisions themselves.

4.2 Assumptions

The model assumes a simplified two-dimensional periodic observation window, constant-speed linear motion between collisions of perfectly spherical objects. Object mass was not modeled in order to maintain the study's focus on positional uncertainty. So, naturally, the logic of acquiring the center of mass velocity upon collision is not present (instead, the average velocity is used). The Gaussian model is assumed to be isotropic instead of elliptical for simplification. Equal, independent uncertainty was therefore assigned to both spatial coordinates. No gravity, atmospheric drag, orbital maneuvers, orbital mechanics or external celestial objects are included in the simulation. Fragmentation followed a simplified size-dependent rule.

Model years represent internal simulation time and are not predictions of real chronological cascade onset. The implemented threshold is an operational definition used to identify cascade onset within the simplified model, rather than a universal physical threshold for real orbital debris environments. Highly enlarged particle radii only served the purpose of allowing collisions to be reasonably probable within the large space and to allow the study of the other effects in the simulation. The simulated population also does not reconstruct the present LEO environment, whose objects are strongly concentrated by altitude. Uniform projection over a representative LEO shell places $N_0 = 10$ within the same order of magnitude for a $600{,}000\ \text{km}^2$ region; however, this is only a scale comparison and not proof of realistic spatial density.

5. RESULTS

A total of 800 runs evaluated the following effects using two initial populations of $N_0 = 6$ and $N_0 = 10$. These were tested at positional uncertainty levels of $\sigma = 0, 1, 2$ and 4 with 100 repetitions under each condition.

Each cascade probability in these figures was calculated from 100 binary Monte Carlo outcomes: a cascade either occurred or did not occur. The plotted 95% error bars were obtained using the Wilson score interval, implemented with the standard $z = 1.96$ value. This method was preferred to the normal approximation, the Wald interval because proportions near 0 or 1 can become unrealistically narrow or extend beyond valid probabilities [9]. Wilson intervals remain physically valid within the range $0\% - 100\%$ and provide a more reliable indication of sampling uncertainty, specifically for this simulation.

5.1 Effect of Tracking Uncertainty

Figure 4 shows the true and observed proportions of simulation runs reaching the generation-3 cascade onset criterion at each tracking-uncertainty level. The horizontal axis represents tracking uncertainty σ, and the vertical axis shows the percentage of runs reaching the criterion. Separate panels illustrate the two studied initial populations. Percentages were calculated using all 100 runs in each condition, including histories in which no cascade onset occurred.

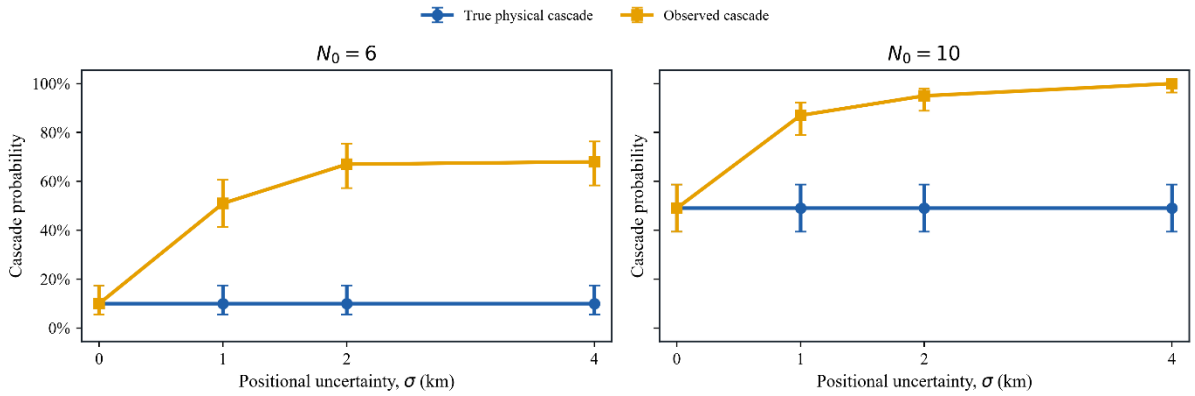


Fig. 4. True (blue) and observed (yellow) cascade probabilities with 95% Wilson confidence intervals across $N_0 = 6$ and $N_0 = 10$. For the smaller initial population ($N_0 = 6$), the observed cascade probability increased from 10% at $\sigma = 0$ km to 51% at $\sigma = 1$ km, 67% at $\sigma = 2$ km, and 68% at $\sigma = 4$ km, while the true probability remained constant at 10%. For the larger population ($N_0 = 10$), the observed probability rose from 49% at $\sigma = 0$ km to 87% at $\sigma = 1$ km, 95% at $\sigma = 2$ km, and 100% (no true-negative outcomes) at $\sigma = 4$ km, compared to a fixed true probability of 49%.

For runs of $\sigma = 4$ km and $N_0 = 10$, every single physical run that remained safe and stable was falsely flagged as a runaway cascade (100% probability). Thus, in such extreme conditions (high σ, N_0), tracking data becomes effectively useless for decision-making. Consequently, tracking uncertainty substantially inflated how frequently generation-3 onset was reported without altering the number of true physical cascades.

5.2 Cascade Onset Misclassifications

Along similar lines, to determine whether the divergence in observed cascade detection arose from false alarms or overlooked events, the classification outcomes were separated into false cascades (false-positive) and missed cascades (false-negative). A false cascade implies an observed onset occurring without a corresponding true onset, whereas a missed cascade denotes a true onset occurring without an observed counterpart.

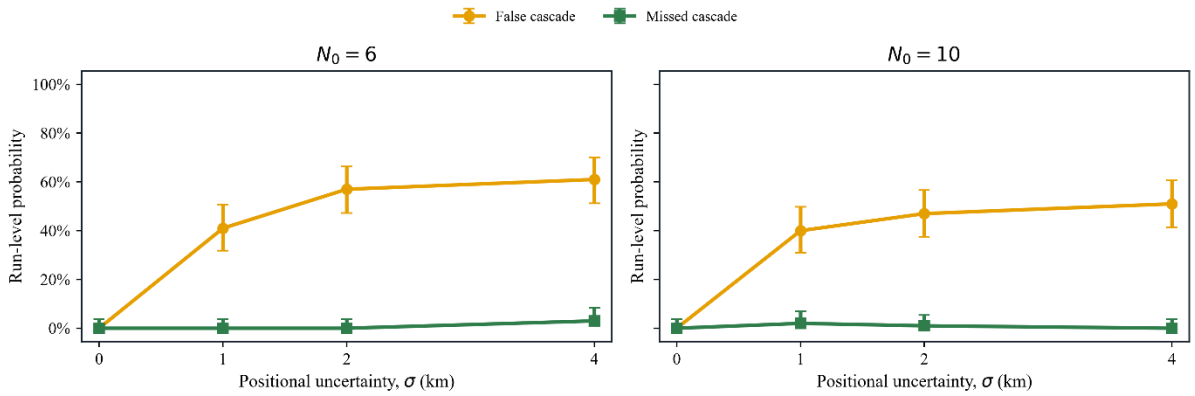


Fig. 5. False (yellow) and missed (green) cascade probabilities with 95% Wilson confidence intervals across $N_0 = 6$ and $N_0 = 10$. At $N_0 = 6$, the false-cascade probability rose from 0% at $\sigma = 0$ km to 41% at $\sigma = 1$ km, 57% at $\sigma = 2$ km, and 61% at $\sigma = 4$ km. At $N_0 = 10$, it increased from 0% to 40%, 47%, and 51% across the same uncertainty steps. Out of 303 total misclassification errors across all 800 runs, 297 were false cascades (98%) and only 6 were missed cascades (2%).

False observed cascade objects therefore overwhelmingly drive the divergence in the system. Tracking uncertainty produces spurious ancestry connections far more frequently than concealing real physical cascades. Greater positional uncertainty generally causes the observed system to cross the generation-3 threshold more often than the physical system, with the effect appearing across both population sizes.

Crucially, these results show that the uncertainty effect was not limited to just accelerating the detection of timing of real physical cascades as one might conclude from Figure 4. Instead, under the tested conditions, observational uncertainty primarily introduced false conjunctions that caused some physically non-cascading systems to cross the generation-3 threshold in the observed layer.

To quantify the reliability of an observed cascade warning, precision and recall were calculated separately for each condition.

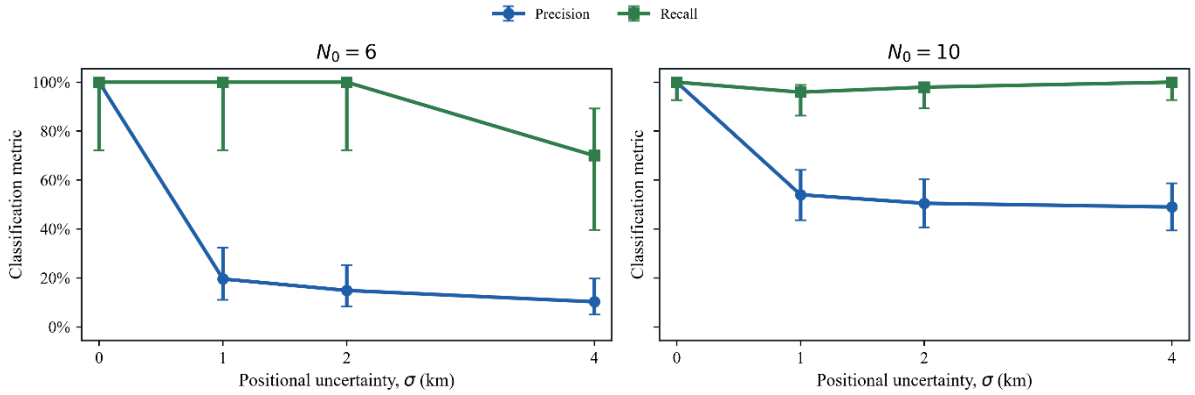


Fig. 6. Precision (blue) and recall (green) of observed cascade detection across tracking uncertainty for $N_0 = 6$ and $N_0 = 10$. Error bars show 95% Wilson confidence intervals. At $\sigma = 0$ km, both metrics were 100% because the true and observed classifications matched. Precision then declined sharply, reaching 10.3% (95% CI: 5.1%–19.8%) for $N_0 = 6$ and 49.0% (95% CI: 39.4%–58.7%) for $N_0 = 10$ at $\sigma = 4$ km.

Recall remained high in most conditions, although it decreased to 70.0% for $N_0 = 6$ at $\sigma = 4$ km. The fall in precision therefore resulted primarily from the growing number of false observed warnings rather than widespread failure to detect true cascades. This was most pronounced for $N_0 = 6$ at $\sigma = 4$ km, where only 7 of 68 observed cascade detections corresponded to true cascades. Observed cascade warnings should therefore be interpreted in the context of the underlying positional tracking uncertainty, since higher uncertainty substantially lowers their reliability.

5.3 Matched Onset-Bias Distributions

When both the true and observed layers register a cascade, tracking uncertainty systematically shifts the detected timing of the onset. These box plots show only runs in which both true and observed generation-3 onset existed. The term "matched" indicates the runs that satisfy both these conditions.

As discussed earlier, there are two relevant cases for onset bias values (B) which are studied. This bias was calculated exclusively for runs in which both true and observed generation-3 onsets existed, preventing false-only and no-onset runs from influencing the timing metrics. Here, for each population, out of 100 runs, the number of true cascade onsets is noted as n i.e. the matched sample size, for zero positional standard deviation. Note that regarding the 6 missed cascades, only T_{true} exists, and not T_{obs}.

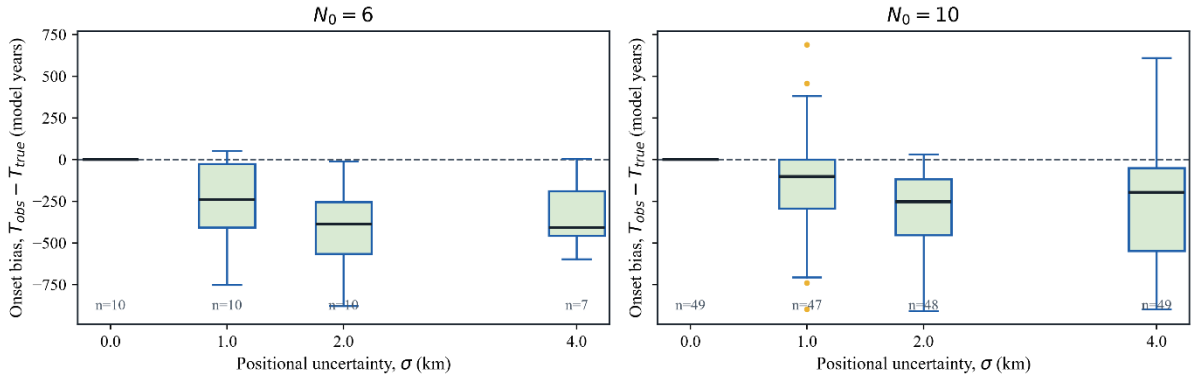


Fig. 7. Matched observed-onset bias B under increasing tracking uncertainty. Negative values indicate earlier observed onset. When $\sigma = 0$ km in both populations (0 missed in both n = 10, n = 49), $B = 0$ because the timing does not vary. At $N_0 = 6$, the median bias for nonzero uncertainty conditions ($\sigma = 1$,2 and 4 km) was -239.4, -386.7, and -406.3 model years, based on the 10 out of 100 runs that experienced a real collision. The matched sample sizes were therefore of 10 (0 missed), 10 (0 missed), and 7 (3 missed) runs respectively at each σ level. At $N_0 = 10$, the median bias was -102.1, -252.0, and -196.7 model years across matched sample sizes of 47 (2 missed), 48 (1 missed), and 49 (0 missed) runs. The median bias remains negative at every nonzero uncertainty level, demonstrating that noisy observations typically assign generation-3 onset earlier than the true physical layer because $B < 0$ follows from $T_{\text{obs}} < T_{\text{true}}$.

Nevertheless, the distributions are broad, showing substantial run-to-run variation and occasional positive biases (around 5-10% of the 171 nonzero σ matched runs), meaning the timing shift is not perfectly monotonic. Still, the onset-bias distribution also became substantially broader at high uncertainty.

Because false-only observed cascades lack a true onset time for comparison, they are absent from these box plots, reinforcing the necessity of interpreting Figure 7 alongside the broader classification outcomes in Figures 4 and 5. Ultimately, within this simplified model, tracking uncertainty primarily generates false observed cascade onsets and causes matched cascades to appear generally earlier, while still being considerably variable in the resulting temporal distortion.

6. CONCLUSION

Rather than merely accelerating the detection of genuine cascades, positional uncertainty primarily caused the model to report cascades that did not physically occur: 297 of 303 classification errors (98%) were false cascade detections. Among runs where both true and observed cascade onsets occurred, the median observed onset occurred 102.1 to 406.3 model years earlier at nonzero uncertainty. Most matched cascades therefore appeared earlier, but the size of this timing shift differed substantially between runs, and a small minority appeared later. Taken together, these findings demonstrate that positional tracking uncertainty can distort both the classification and apparent timing of cascade onset within the present model.

This distinction is relevant to SSA, where conjunction and collision risk assessment depend on predicted object positions and their associated uncertainty [10]. The results therefore support the use of uncertainty-aware interpretation of apparent cascade risk trends. For long-term space sustainability, improved tracking accuracy and regular updating of orbital data can help distinguish an actual increase in collision risk from an observation-driven warning, supporting more reliable collision avoidance and debris mitigation decisions [10], [11].

7. REFERENCES

[1] C. J. Liang, P. E. Fanto, and A. J. Signoracci, "On the risk of Kessler syndrome: A statistical modeling framework for orbital debris growth," *J. Astronaut. Sci.*, vol. 71, no. 5, Art. no. 45, 2024, doi: 10.1007/s40295-024-00458-3.

[2] A. Rao and F. Letizia, "An integrated debris environment assessment model," in *Proc. 8th Eur. Conf. Space Debris*, Darmstadt, Germany, 2021. Accessed: Aug. 9, 2026. [Online]. Available: https://conference.sdo.esoc.esa.int/proceedings/sdc8/paper/206/SDC8-paper206.pdf

[3] S. Nikolaev *et al*., "Brute force modeling of the Kessler syndrome," in *Proc. Advanced Maui Optical and Space Surveillance Technologies Conf. (AMOS)*, Maui, HI, USA, 2012. Accessed: Aug. 9, 2026. [Online]. Available: https://amostech.com/TechnicalPapers/2012/Orbital_Debris/NIKOLAEV.pdf

[4] H. G. Lewis, "Understanding long-term orbital debris population dynamics," *J. Space Saf. Eng.*, vol. 7, no. 3, pp. 164–170, 2020, doi: 10.1016/j.jsse.2020.06.006.

[5] C. Greco, L. Sánchez, M. Manzi, and M. Vasile, "A robust Bayesian agent for optimal collision avoidance manoeuvre planning," in *Proc. 8th Eur. Conf. Space Debris*, Darmstadt, Germany, 2021. Accessed: Aug. 9, 2026. [Online]. Available: https://strathprints.strath.ac.uk/77591/1/Greco_etal_ECSD_2021_A_robust_Bayesian_agent_for_optimal_collision_avoidance.pdf

[6] E. Blasch, D. Shen, G. Chen, C. Sheaff, and K. Pham, "Space object tracking uncertainty analysis with the URREF ontology," in *2021 IEEE Aerosp. Conf.*, 2021, pp. 1–9, doi: 10.1109/AERO50100.2021.9438207.

[7] B. A. Jones, A. Doostan, and G. H. Born, "Conjunction assessment using polynomial chaos expansions," in *Proc. 23rd Int. Symp. Space Flight Dynamics*, Pasadena, CA, USA, 2012, pp. 1–18. Accessed: Aug. 9, 2026. [Online]. Available: https://www.issfd.org/ISSFD_2012/ISSFD23_CRSD2_1.pdf

[8] European Space Agency, "Low Earth orbit," ESA, Mar. 2, 2020. Accessed: Aug. 9, 2026. [Online]. Available: https://www.esa.int/ESA_Multimedia/Images/2020/03/Low_Earth_orbit

[9] L. D. Brown, T. T. Cai, and A. DasGupta, "Interval estimation for a binomial proportion," *Stat. Sci.*, vol. 16, no. 2, pp. 101–133, 2001, doi: 10.1214/ss/1009213286.

[10] F. J. Krage *et al*., *NASA Spacecraft Conjunction Assessment and Collision Avoidance Best Practices Handbook*, Rev. 1, NASA/SP-20230002470, National Aeronautics and Space Administration, Washington, DC, USA, 2023. Accessed: Aug. 9, 2026. [Online]. Available: https://ntrs.nasa.gov/citations/20230002470

[11] European Space Agency, "Mitigating space debris generation," ESA. Accessed: Aug. 9, 2026. [Online]. Available: https://www.esa.int/Space_Safety/Space_Debris/Mitigating_space_debris_generation

APPENDIX

Table A1 provides the principal numerical settings of the Monte Carlo experiment. Together, these values establish the model's spatial scale and duration, the tested uncertainty conditions, and the criterion used to identify cascade onset.

Table A1. Principal parameters used in the experiment.

Quantity	Symbol	Value
Window width	L_x	1000 km
Window height	L_y	600 km
Initial population	N_0	6 and 10 objects
Time step	Δt	0.1 model years
Simulation duration	T_{max}	1000 model years
Repetitions per condition	n_{rep}	100
Initial object radius	R_0	1.2 km
Initial speed	s_i	7-8 km s^{-1}
Fragment radius	R_f	150-350 m
Positional uncertainty	σ	0, 1, 2, 4 km
Cascade onset threshold	g	Generation 3 ($g \geq 3$)

The following equations specify the initialization and periodic-boundary implementation used in the simulation. Initial object positions were sampled uniformly within the observation window, while velocity directions were sampled uniformly over $0 \leq \theta_i < 2\pi$. Periodic boundary updates ensured that objects leaving the simulation window re-entered from the corresponding opposite boundary.

Initial position, velocity vectors and periodic boundary position updates of object i were respectively:

$$\vec{r}_i(0) = \begin{bmatrix} X_i \\ Y_i \end{bmatrix}, \qquad X_i \sim \mathcal{U}(0, L_x), \qquad Y_i \sim \mathcal{U}(0, L_y) \tag{A1}$$

$$\vec{v}_i = \begin{bmatrix} v_i \cos\theta_i \\ v_i \sin\theta_i \end{bmatrix}, \qquad \theta_i \sim \mathcal{U}(0, 2\pi) \tag{A2}$$

$$\vec{r}_{i,\text{boundary}} = \begin{bmatrix} r_{i,x} \mod L_x \\ r_{i,y} \mod L_y \end{bmatrix} \tag{A3}$$

Figure A1 shows a visual-mode snapshot of the 1000×600 km periodic observation window for a test with $N_0 = 6$. It displays labelled object locations at a single model time. All Monte Carlo analyses reported in this study were instead executed in headless mode.

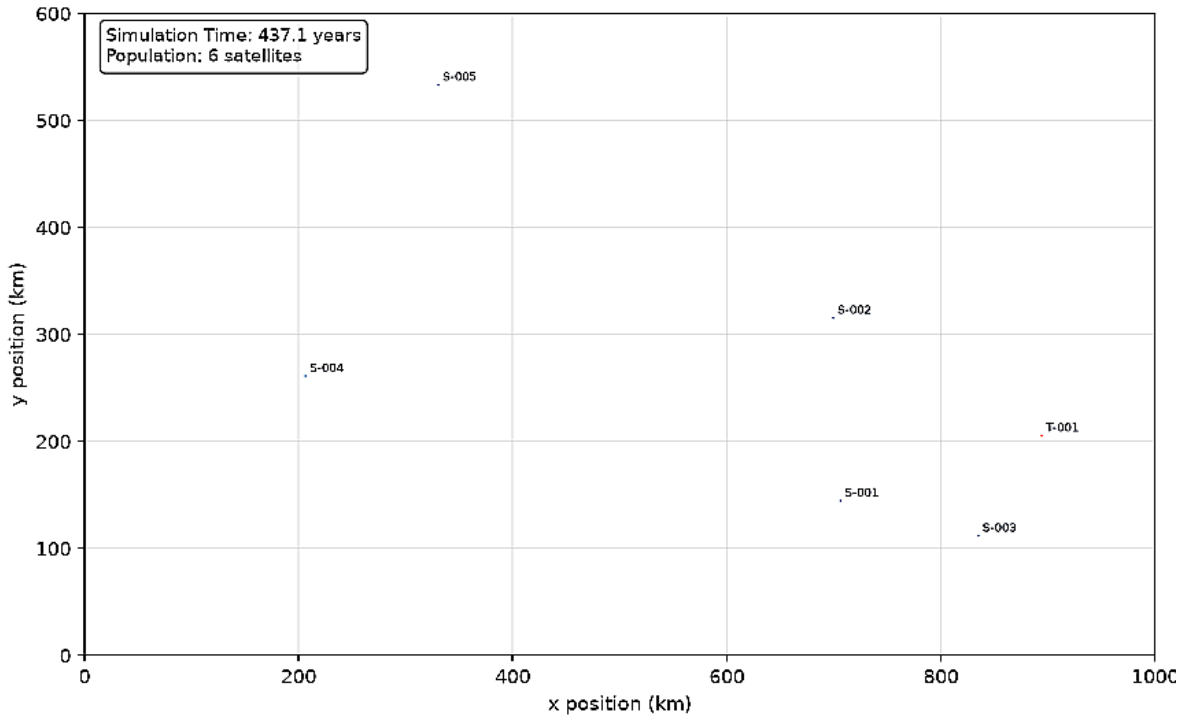


Fig. A1. Visualization of a single simulation state within the periodic observation window.